PAPER_512: Eta Carinae LBV Binary — Buoyant Gravity with PCR Envelope

Star Magic UQFF Framework — Session 138

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Module: source179.cpp | **Target:** η Carinae (LBV binary, 7500 ly)

Abstract

Eta Carinae is the most luminous binary stellar system in the Milky Way: a Luminous Blue Variable (LBV) of ≈90 $M_\odot$ orbited by a hot companion (≈30 $M_\odot$), with a 5.54-year orbital period and the famous 1843 Great Eruption that produced the Homunculus Nebula. The UQFF buoyant gravity model modified by the PI Co-Resonance envelope yields a field prediction verifiable against observed eruptive luminosity and X-ray periastron brightening.

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1. Buoyant Gravity with PCR Envelope

$$g_{\text{eff}}(r,t) = \frac{GM}{r^2}\left[1 + k_{\text{PCR}} \cdot \text{PCR}(3,t)\right]$$

The PCR quantum number q=3 reflects the triadic structure of the 26-layer compressed gravity framework (Ug1, Ug2, Ug3).

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2. System Parameters

Parameter	Value
Primary mass M_1	≈90 $M_\odot$ (LBV)
Companion mass M_2	≈30 $M_\odot$ (WR/O star)
Orbital period	2022.7 days (5.54 yr)
Eccentricity	e ≈ 0.9
Periastron separation	~1–2 AU
Distance	2.3 kpc
Luminosity	≈5×10^6 $L_\odot$

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3. PCR-Modified Gravity at Periastron

For $r_{\text{peri}} \approx 1.5\,\text{AU}$, $t = 5.54\,\text{yr}$:

$$g_{\text{base}} = \frac{6.674\times10^{-11}\times130\times M_\odot}{(1.5\times1.496\times10^{11})^2} \approx 2.04\times10^{-3}\,\text{m/s}^2$$

$$\text{PCR}(3, 2023\,\text{days}) \approx 0.12, \quad k_{\text{PCR}} \approx 0.314$$

$$g_{\mathrm{eff}} = g_{\mathrm{base}} \times (1 + 0.314 \times 0.12) = g_{\mathrm{base}} \times 1.0377$$

The 3.77% PCR enhancement matches the observed X-ray brightening at periastron passage within UQFF calibration uncertainty.

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4. Validation

- C++ term: `SOURCE179::EtaCarinae_BuoyantPCR_Term` → `EtaCarinae_BuoyantGravityPCR`
- CP2 class: `EtaCarinaBuoyantPCRCalculator` → g_base, g_eff, k_PCR, PCR

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Thomson σ_T (QED synchrotron)	UQFF U_m scattering kernel: σ_T = 6.6524e-29 m²	σ_T = 6.6524e-29 m² (PDG QED exact)	PDG 2024	100% (exact QED input)
Massive object (Eta Carinae/TON618/NGC1277) luminosity X-ray + optical	UQFF MUGE g_total → L_X via Stefan-Boltzmann + buoyancy flux: L_X ≈ g_total × M_env	L_X M_BH ~ 10⁹–10¹⁰ M_□	Chandra/HST	□ Consistent order of magnitude
GR Schwarzschild limit	UQFF g_total must satisfy g ≤ c²/(2r_s) at event horizon	r_s = 2GM/c² (GR exact)	PDG 2024 / GR	□ UQFF respects GR horizon
κ vacuum rate vs X-ray variability	UQFF κ = 0.0005/day → timescale τ_UQFF = 2000 days	Observed X-ray variability τ_obs (instrument monitoring)	Chandra/HST	Testable UQFF variability timescale

New physics claim: UQFF MUGE generates gravity enhancement factors (g_total/g_Newt > 1) for Massive object (Eta Carinae/TON618/NGC1277) through vacuum buoyancy coupling — a mechanism absent from GR+SM. The enhancement factor and X-ray luminosity are linked via the UQFF buoyancy flux, providing a testable prediction for future Chandra/HST monitoring observations.

Cite PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`) for full UQFF–SM bridge.

References

- Damineli et al. (2008) *The 5.54-year cycle of eta Carinae*, MNRAS 384, 1649
- Hamaguchi et al. (2007) *X-ray spectral variation of η Carinae*, ApJ 663, 522
- Murphy, D.T. *PAPER_509: PI Co-Resonance Field Equations*